Kyle Nwogu

CHE 312 Transport Phenomena II
Instructor Vivek Sharma

Final: Manufacturing Pharmaceuticals using concepts from Transport Phenomena II

(Includes original problems: do not repost anywhere without permission)
Marked out of 100; contribution to the final grade varies
Edited Appendix B from the Transport Phenomena text (by BSL) is available for reference. Handwritten formula sheet will be graded too and is worth five points (at maximum). Assume that the time in minutes needed for solving a problem is comparable to (or less than) the maximum score possible on that problem.

Problem 1: For a given concentration profile (determined by an chemical engineer intern at a pharmaceutical company), $\frac{c}{c_0} = \sin\left(-\pi x^2/4D_{AB}t\right) - \left(\frac{x^2}{4D_{AB}t}\right)\left(1 - \text{erf}\frac{x}{\sqrt{4D_{AB}t}}\right)$, where c_0 and D_{AB} are both constant and known **(6+4+5=15)**, find:
(a) the local mass flux in x-direction at $x = 0$.
(b) the concentration at x = 0 and at $x^2 = 4D_{AB}t$.
(c) $\frac{Dc}{Dt}$.

$C = C(x) \neq C(y) = C(z) = 0$

a) $C = C_0\left(\sin(-\pi x^2/4D_{AB}t) - \left(\frac{x^2}{4D_{AB}t}\right)\left(1 - \text{erf}\frac{x}{\sqrt{4D_{AB}t}}\right)\right)$

$\frac{dc}{dx} = C_0\left(\frac{-2\pi x}{4D_{AB}t}\cos(-\pi x^2/4D_{AB}t) - \frac{2x}{4D_{AB}t} + \frac{2x}{4D_{AB}t}\text{erf}\frac{x}{\sqrt{4D_{AB}t}} + \frac{x^2}{4D_{AB}t}\cdot\frac{2}{\sqrt{\pi}}\exp\left(-\frac{x^2}{4D_{AB}t}\right)\cdot\frac{1}{\sqrt{4D_{AB}t}}\right)$

$\left.\frac{dc}{dx}\right|_{x=0} = C_0(0) = 0$

$N_{Ax}|_{x=0} = -D_{AB}\left.\frac{dc}{dx}\right|_{x=0} = -D_{AB}\cdot 0 = 0$ //

b) at $x = 0$

$C = C_0\left(\sin(0) - \left(\frac{0^2}{4D_{AB}t}\right)\left(1 - \text{erf}\frac{0}{\sqrt{4D_{AB}t}}\right)\right)$

$\Rightarrow C = 0$ //

at $x^2 = 4D_{AB}t$

$\text{erf}(1) = 0$

$C = C_0\left(\sin\left(\frac{-\pi\, \cancel{4D_{AB}t}}{\cancel{4D_{AB}t}}\right) - \left(\frac{\cancel{4D_{AB}t}}{\cancel{4D_{AB}t}}\right)\left(1 - \cancel{\text{erf}\frac{\sqrt{4D_{AB}t}}{\sqrt{4D_{AB}t}}}\right)\right)$

$C = C_0\left(\overset{0}{\cancel{\sin(-\pi)}} - 1\right)$

$C = -C_0$ //

c) $\frac{Dc}{Dt} = \cancel{\frac{dc}{dt}} + V_x\frac{dc}{dx} + \cancel{V_y\frac{dc}{dy}} + \cancel{V_z\frac{dc}{dz}}$

$\frac{Dc}{Dt} = V_x\cdot C_0\left(\frac{-2\pi x}{4D_{AB}t}\cos\left(\frac{-\pi x^2}{4D_{AB}t}\right) - \frac{2x}{4D_{AB}t} + \frac{2x}{4D_{AB}t}\text{erf}\frac{x}{\sqrt{4D_{AB}t}} + \frac{x^2}{4D_{AB}t}\cdot\frac{2}{\sqrt{\pi}}\exp\left(-\frac{x^2}{4D_{AB}t}\right)\cdot\frac{1}{\sqrt{4D_{AB}t}}\right)$ //

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Problem 2: Heat Transfer and Fever: Designer Cold Patches
The forehead of a person running a high fever is at temperature T_0. A cold wet towel of thickness d and initial temperature $T_1 < T_0$ is placed on the forehead at time $t = 0$. **Assuming the body-temperature is maintained constant, find the time-dependent temperature profile within the towel. Also determine the heat flux into the towel. For a wet towel of area A, how much heat is transferred per second from the body to the towel?** (You can probably help a pharmaceutical company design more efficient cold patches using a better understanding of heat transfer concepts involved, and the analysis you write down here can be used for picking the right material or right thickness for wet towels). **(12+4+4)**

$$\rho \hat{C}_p\left(\frac{\partial T}{\partial t} + \cancel{v_x \frac{\partial T}{\partial x}} + \cancel{v_y \frac{\partial T}{\partial y}} + \cancel{v_z \frac{\partial T}{\partial z}}\right) = k\left[\cancel{\frac{\partial^2 T}{\partial x^2}} + \frac{\partial^2 T}{\partial y^2} + \cancel{\frac{\partial^2 T}{\partial z^2}}\right] + \cancel{\mu \Phi_v}$$

$$\Rightarrow \frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial y^2}$$

B.C: $y=0$, $T=T_1$ for all $t>0$

$y \to \infty$, $T = T_0$, for all $t>0$

Introduce dimensionless temp $\Theta = \dfrac{T-T_0}{T_1-T_0}$

$$\Rightarrow \frac{\partial \Theta}{\partial t} = \alpha \frac{\partial^2 \Theta}{\partial y^2}$$

$$\eta = y/\sqrt{4\alpha t}$$

comb of variables w/ respect to η

$$\frac{d\Theta}{dt} = \frac{d\Theta}{d\eta}\frac{d\eta}{dt} \quad \Big\| \quad \frac{d\eta}{dt} = \frac{d}{dt}\left(t^{-1/2}\right)\cdot\frac{y}{\sqrt{4\alpha}} = -\frac{1}{2}\frac{y}{\sqrt{4\alpha}}t^{-3/2} = -\frac{1}{2}\eta/t$$

$$\frac{d\Theta}{dy} = \frac{d\Theta}{d\eta}\frac{d\eta}{dy} \quad \Big\| \quad \frac{d\eta}{dy} = \frac{d}{dy}(y)\frac{1}{\sqrt{4\alpha t}} = \frac{1}{\sqrt{4\alpha t}} = \eta/y \quad \Rightarrow \frac{d^2\Theta}{dy^2} = \frac{d^2\Theta}{d\eta^2}\left(\eta/y\right)^2$$

$$-\frac{1}{2}\eta/t\,\frac{d\Theta}{d\eta} = \cancel{\alpha}\frac{\partial^2\Theta}{\partial\eta^2}\cdot\frac{1}{4\alpha t}$$

$$\Rightarrow -2\eta\frac{d\Theta}{d\eta} = \frac{\partial^2\Theta}{\partial\eta^2} \Rightarrow \frac{\partial^2\Theta}{\partial\eta^2} + 2\eta\frac{d\Theta}{d\eta} = 0$$

B.C: $\eta = 0$, $\Theta = 1$

$\eta \to \infty$, $\Theta = 0$

we see ODE is in form of C.1-8, hence

$$\Theta = C_1\int_0^\eta \exp(-\bar{\eta}^2)\,d\bar{\eta} + C_2$$

@ B.C#1,

$$1 = C_1\cancel{\int_0^0 \exp(-\bar{\eta}^2)}\,d\eta + C_2$$

$C_2 = 1$

at B.C#2: $0 = C_1\int_0^\infty \exp(-\bar{\eta}^2)\,d\bar{\eta} + 1$

$$C_1 = \frac{-1}{\int_0^\infty \exp(-\bar{\eta}^2)\,d\bar{\eta}}$$

$$\Theta = 1 - \underbrace{\frac{\int_0^\eta \exp(-\bar{\eta}^2)\,d\bar{\eta}}{\int_0^\infty \exp(-\bar{\eta}^2)\,d\bar{\eta}}}_{\text{erf}\,\eta}$$

$$\Rightarrow \frac{T-T_0}{T_1-T_0} = 1 - \text{erf}\; y/\sqrt{4\alpha t}$$

$$T = (T_1-T_0) - (T_1-T_0)\,\text{erf}\frac{y}{\sqrt{4\alpha t}}$$

$$\frac{dT}{dy} = -(T_1-T_0)\cdot\frac{2}{\sqrt{\pi}}\exp\left(-\frac{y^2}{4\alpha t}\right)\cdot\frac{1}{\sqrt{4\alpha t}}$$

$$\frac{dT}{dy}\Big|_{y=0} = -(T_1-T_0)\cdot\frac{1}{\sqrt{\pi\alpha t}}$$

heat flux

$$q|_{y=0} = -k\frac{dT}{dy}\Big|_{y=0} = -k\left(-(T_1-T_0)\frac{1}{\sqrt{\pi\alpha t}}\right) = \frac{k}{\sqrt{\pi\alpha t}}(T_1-T_0)$$

heat transferred

$$Q|_{y=0} = A\cdot q|_{y=0} = A\cdot\frac{k}{\sqrt{\pi\alpha t}}(T_1-T_0)$$

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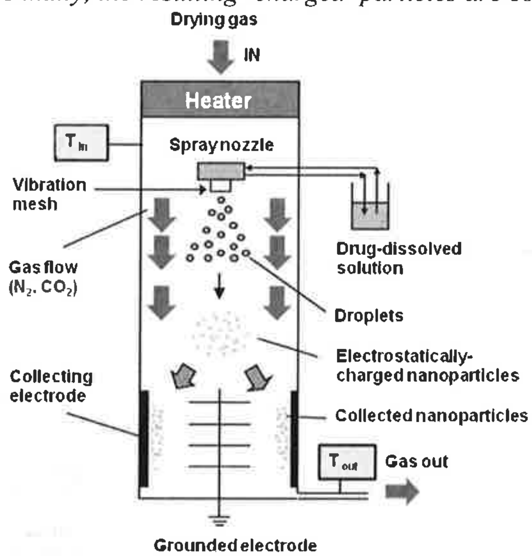
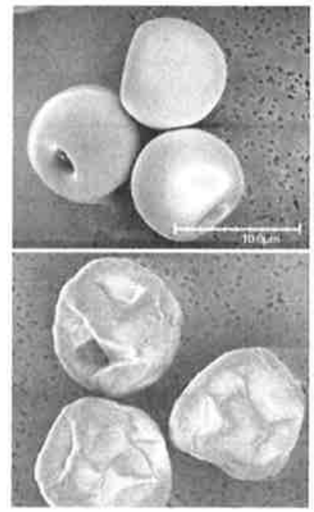
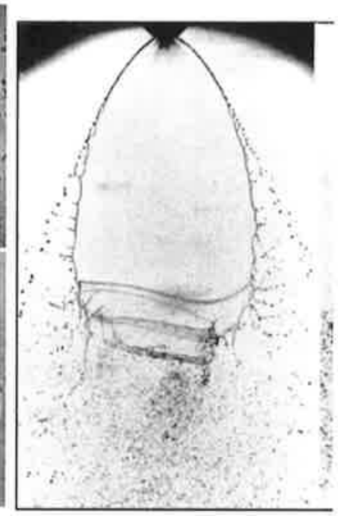
Problem 3. Spray-dried Pharmaceuticals: *Spray-drying process (see the schematic below) is commonly used in the pharmaceutical and food industries for making particles or powders (including dry milk, flavored coffee, etc.). In a typical process, the drug-dissolved solution is through a spray nozzle. The fluid atomizes as it issues out of the nozzle, and the resulting droplets dry out in presence of a drying gas. Finally, the resulting 'charged' particles are collected as shown.* **(25+25+10=60)**







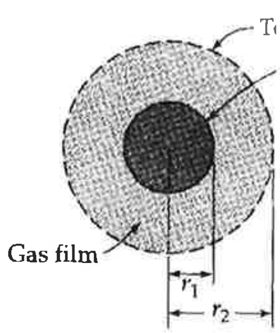
(Figures credits: *Pharm. Res.* 25(5), 2008 and Villermaux, *Ann. Rev. Fluid Mech.* 2007)

(a) Assume that a hypothetical spherical stagnant film of gas B surrounds each falling droplet made of fluid A. Assume that a steady-state diffusion of A occurs through the stagnant film of gas B, and the solubility of gas B in liquid A is negligible. Assume that the gas-phase concentration of A, expressed as mole fraction is x_{A1} at the liquid-gas interface (r_1) and the concentration just outside the stagnant film is maintained at x_{A2}. Assume gases A and B are ideal and assume that change in drop size can be neglected. Derive the expressions for diffusion through a spherical shell to obtain both the concentration profile and molar flux. (25)

Shell Balance: $4\pi r^2 N_{Ar}|_r - 4\pi r^2 N_{Ar}|_{r+\Delta r} = 0$

B.C stagnant gas film B, $N_{Br}=0$

divide by $4\pi r^2 \Delta r$ & take limit as $\Delta r \to 0$



$$\lim_{\Delta r \to 0} \frac{N_{Ar}|_r - N_{Ar}|_{r+\Delta r}}{\Delta r} = 0 \Rightarrow -\frac{d(N_{Ar})}{dr} = 0$$

Flux relationship: $N_{Ar} = x_A(N_{Ar} + \cancel{N_{Br}}^{\,0}) - c D_{AB}\frac{dx_A}{dr}$

$$N_{Ar} - x_A N_{Ar} = -c D_{AB}\frac{dx_A}{dr}$$

$$\Rightarrow N_{Ar} = \frac{1}{1-x_A}\cdot -c D_{AB}\frac{dx_A}{dr}$$

$$\frac{d}{dr}\left(r^2 \frac{1}{1-x_A}\cdot c D_{AB}\frac{dx_A}{dr}\right) = 0$$

if P & T const ⇒ c is constant, D_{AB} is const. b/c ind of concentration for gases

$$\frac{d}{dr}\left(\frac{1}{1-x_A}\frac{dx_A}{dr}\right) = 0$$

B.C: $r = r_1,\ x_A = x_{A1}$

$r = r_2,\ x_A = x_{A2}$

$$\int d\left(\frac{1}{1-x_A}\frac{dx_A}{dr}\right)=\int 0\,dr$$

$$\frac{1}{1-x_A}\frac{dx_A}{dr}=C_1$$

$$\Rightarrow \frac{1}{1-x_A}dx_A = C_1\,dr$$

$$\Rightarrow \int \frac{1}{1-x_A}dx_A = \int C_1\,dr$$

$$-\ln(1-x_A) = C_1 r + C_2$$

at B.C #1, $r=r_1$ $x_A=x_{A1}$

$$-\ln(1-x_{A1}) = C_1 r_1 + C_2$$

at B.C #2, $r=r_2$, $x_A=x_{A2}$

$$-\ln(1-x_{A2}) = C_1 r_2 + C_2$$

combine eq.'s to solve for C_1

$$-\ln(1-x_{A1}) = C_1 r_1 + C_2$$

$$-\left(-\ln(1-x_{A2}) = C_1 r_2 + C_2\right)$$

$$-\ln(1-x_{A1}) + \ln(1-x_{A2}) = C_1 r_1 - C_1 r_2$$

$$-\ln(1-x_{A1}) + \ln(1-x_{A2}) = C_1(r_1-r_2)$$

$$C_1 = -\ln(1-x_{A1})^{1/(r_1-r_2)} + \ln(1-x_{A2})^{1/(r_1-r_2)}$$

$$-\ln(1-x_{A1}) = -\ln(1-x_{A1})^{r_1/r_1-r_2} + \ln(1-x_{A2})^{r_1/r_1-r_2} + C_2$$

$$C_2 = -\ln(1-x_{A1}) + \ln(1-x_{A1})^{r_1/r_1-r_2} - \ln(1-x_{A2})^{r_1/r_1+r_2}$$

$$-\ln(1-x_A) = +\ln(1-x_{A1})^{r/r_2-r_1} - \ln(1-x_{A2})^{r/r_2-r_1} + \ln(1-x_{A1}) - \ln(1-x_{A1})^{r_1/r_2-r_1} + \ln(1-x_{A2})^{r_1/r_2-r_1}$$

Mult. by -1

$$\ln(1-x_A) = -\ln(1-x_{A1})^{r/r_2-r_1} + \ln(1-x_{A2})^{r/r_2-r_1} + \ln(1-x_{A1}) + \ln(1-x_{A1})^{r_1/r_2-r_1} - \ln(1-x_{A2})^{r_1/r_2-r_1}$$

$$\ln(1-x_A) - \ln(1-x_{A1}) = \ln\left(\frac{1-x_{A2}}{1-x_{A1}}\right)^{\frac{r-r_1}{r_2-r_1}}$$

$$\Rightarrow \left(\frac{1-x_A}{1-x_{A1}}\right) = \left(\frac{1-x_{A2}}{1-x_{A1}}\right)^{\frac{r-r_1}{r_2-r_1}}$$

Flux

$$N_{Ar} = -\frac{cD_{AB}}{1-x_A}\frac{dx_A}{dr}$$

$$N_{Ar}\int_{r_1}^{r_2} dr = -cD_{AB}\int_{x_{A1}}^{x_{A2}}\frac{1}{1-x_A}dx_A$$

$$N_{Ar}(r_2-r_1) = -cD_{AB}\cdot -\ln\left(\frac{1-x_{A2}}{1-x_{A1}}\right)$$

$$\Rightarrow N_{Ar} = \frac{cD_{AB}}{(r_2-r_1)}\ln\left(\frac{1-x_{A2}}{1-x_{A1}}\right)$$

(b) **Freezing Drops**. Rather than drying, some pharmaceutical are produced by freezing the droplets. In such a scenario, the droplets leave the nozzle only slightly above their melting point, and drying gas plays the role of a colder outside medium. Estimate the time t_f required for a drop of radius R to freeze completely, if the drop is initially at its melting point T_0 and the surrounding air is at T_1. Heat is lost from the drop according to the Newton's law of cooling, with constant heat-transfer coefficient h. Assume that there is no volume change in the solidification process. Solve the problem by using a **quasi-steady-state method**.

(HINT: (a) First solve the steady-state heat conduction problem in the solid phase in the region between $r = R_f$ (the liquid-solid interface) and $r = R$ (the solid-air interface). Let k be the thermal conductivity of the solid phase. Then find the radial heat flow Q across the spherical surface at $r = R$. (b) Assuming H be the latent heat of freezing per unit mass, and using unsteady-state energy balance, find the time that drop takes to solidify). (25=15+10)

B.C.: $r = R_f$, $T = T_0$

$r = R$, $T = T_\infty$

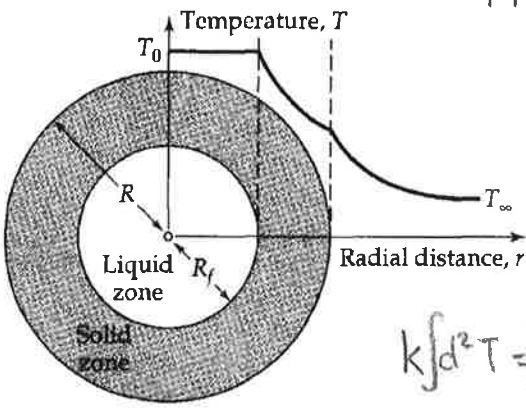
a) $A q_r|_r - A q_r|_{r+\Delta r} = 0$

divide by $A\Delta r$ & take limit as $\Delta r \to 0$

$\Rightarrow \dfrac{-dq_r}{dr} = 0$

insert flux law, $q_r = -k\, dT/dr$

$\Rightarrow k \dfrac{d^2T}{dr^2} = 0$



$k\int d^2T = \int 0\, dr^2$

$\int k\, dT = \int C_1\, dr$

$kT = C_1 r + C_2$

at B.C #1,

$kT_0 = C_1 R_f + C_2$

at BC#2

$kT_\infty = C_1 R + C_2$

combine

$\Rightarrow kT_0 - kT_\infty = C_1 R_f - C_1 R$

$\Rightarrow \dfrac{k(T_0 - T_\infty)}{(R_f - R)} = C_1$

plug back in,

$kT_0 = \dfrac{k(T_0 - T_\infty)}{(R_f - R)} R_f + C_2$

$\Rightarrow C_2 = kT_0 - \dfrac{k(T_0 - T_\infty)}{(R_f - R)} R_f$

$kT = k\left(\dfrac{(T_0 - T_\infty)}{(R_f - R)} R_f + T_0 - \dfrac{(T_0 - T_\infty)}{(R_f - R)} R_f\right)$

(c) Mass transfer from a droplet: (All quantities in this problem are in SI units. Appropriate units must accompany your final numerical answer). The droplets of diameter of 0.01 (in SI units) fall through the gas B with velocity of 4 SI units. The drying gas has viscosity 4 x 10^{-4} units, thermal conductivity of 0.025, density of 1.1, specific heat per unit mass of 1000 and the diffusion coefficient is $1/10^9$ SI units. **Give a simple estimate for mass transfer coefficient relevant for this problem, useful for estimating mass flux out of drying droplets.** (10)

Flux [=] $kg/m^2 \cdot s$

$Pa = N \cdot m = kg\, m^2/s$

$D = .01 m$

$V = 4 m/s$

$\mu = 4 \times 10^{-4}\ Pa \cdot S$

$k = .025\ W/m \cdot K$

$\rho = 1.1\ kg/m^3$

$\hat{C}_p = 1000 J/kg \cdot K$

$D_{AB} = \frac{1}{10^9}\ m^2/s$

$D \cdot V = .01 \times 4 = .04\ m^2/s$

$\frac{k}{\hat{C}_p} = \frac{.025 \frac{J}{m \cdot s \cdot K}}{1000 J/kg \cdot K} = 2.5 \times 10^{-5}\ kg/m \cdot s$

$\frac{\left(\frac{k}{\hat{C}_p}\right)}{D} = \frac{2.5 \times 10^{-5}\ kg/m \cdot s}{.01 m} = .0025\ kg/m^2 \cdot s$ //

On Heroic Effort

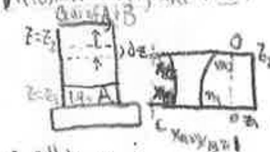
A hero ventures forth from the world of common day into a region of supernatural wonder: fabulous forces are there encountered and a decisive victory is won: the hero comes back from this mysterious adventure with the power to bestow boons on his fellow man.

Joseph Campbell: A Hero with a Thousand Faces.

Kyle Nwagu

$\zeta = \frac{z - z_1}{z_2 - z_1}$

Diffusion in a Stagnant Film:



Assumptions: P, T const for entire system; no radial velocity dependence for escaping gas; liq. A evaporates to gas B; Liq. level maintained at $z = z_1$; Stagnant film B $\Rightarrow N_{Bz} = 0$

B.C #1: $z = z_1$, $x_A = x_{A1}$

B.C #2: $z = z_2$, $x_A = x_{A2}$

Shell Balance: $\frac{dN_{Az}}{dz} = 0$ IN - OUT + PROD = ACCUM

Flux relationship: $N_{Az} = x_A(N_{Az} + N_{Bz}) - cD_{AB}\frac{dx_A}{dz}$

$\Rightarrow \frac{d}{dz}\left(\frac{1}{1-x_A}\frac{dx_A}{dz}\right) = 0$

conc. profile: $\left(\frac{1-x_A}{1-x_{A1}}\right) = \left(\frac{1-x_{A2}}{1-x_{A1}}\right)^{\frac{z-z_1}{z_2-z_1}}$ or $\left(\frac{x_B}{x_{B1}}\right) = \left(\frac{x_{B2}}{x_{B1}}\right)^{z-z_1/z_2-z_1}$

avg. conc: $\frac{x_{B,avg}}{x_{B1}} = \int_{z_1}^{z_2}\left(x_B/x_{B1}\right)dz$

$u = x_{B2}^{3} x_{B1}^{-3}$

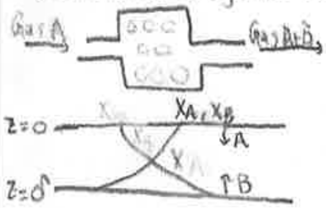
$du = \frac{\ln(x_{B2}) x_{B2}^3}{x_{B1}^3} - \frac{\ln(x_{B1}) x_{B2}^3}{x_{B1}^3}\int_{z_1}^{z_2} dz$

Flux: $N_{Az}|_{z=z_1} = \frac{-cD_{AB}}{1-x_{A1}}\frac{dx_A}{dz}\Big|_{z=z_1} = \frac{cD_{AB}}{x_{B1}}\frac{dx_B}{dz}\Big|_{z=z_1}$

$x_{B,avg} = \frac{x_{B2} - x_{B1}}{\ln(x_{B2}/x_{B1})} = x_{B_{ln}}$ $\Leftarrow$ $\frac{x_{B,avg}}{x_{B1}} = \left(\frac{x_{B2}}{x_{B1}} - 1\right)\Big/\ln\left(x_{B2}/x_{B1}\right)$

$\frac{d}{dx}a^x = \ln a \cdot a^x$

Diffusion w/ a Heterogeneous rxn



Given: $2A \to B$; Thickness of stagnant film layer δ; isoT, steady state; main stream conc: x_{A0}, x_{B0}

From stoichiometry: $N_{Bz} = -\frac{N_{Az}}{2}$ $(2A \to B)$

Shell Balance: $\frac{dN_{Az}}{dz} = 0$ $(A \to 2B)$ $N_{Bz} = -2N_{Az}$

Flux relationship: $N_{Az} = x_A(N_{Az} + N_{Bz}) - cD_{AB}\frac{dx_A}{dz}$

$\Rightarrow \frac{d}{dz}\left(\frac{1}{1-\frac{1}{2}x_A}\frac{dx_A}{dz}\right) = 0$, B.C #1: $z = 0$, $x_A = x_{A0}$; B.C #2: $z = \delta$, $x_A = 0$

Conc. profile: $(1 - \tfrac{1}{2}x_A) = (1 - \tfrac{1}{2}x_{A0})^{1 - z/\delta}$

Flux: $N_{Az}|_{z=0} = \frac{1}{1-\frac{1}{2}x_{A0}} \cdot -cD_{AB}\frac{dx_A}{dz}\Big|_{z=0} = \frac{2cD_{AB}}{\delta}\ln\left(\frac{1}{1-\frac{1}{2}x_{A0}}\right)$

Diffusion w/ Slow Heterogeneous rxn

only difference from Heterogeneous rxn is B.C #2

B.C #2: $z = \delta$, $x_A = N_{Az}/k_1''c$

$N_{Az} = k_1''c_A = k_1''c\,x_A$

Conc profile: $(1 - \tfrac{1}{2}x_A) = \left(1 - \tfrac{1}{2}\left(\frac{N_{Az}}{k_1''c}\right)\right)^{z/\delta} \cdot \left(1 - \tfrac{1}{2}x_{A0}\right)^{1 - z/\delta}$

$N_{Az}|_{z=0} = -cD_{AB}\frac{dx_A}{dz}\Big|_{z=0} \cdot \frac{1}{(1 - \frac{1}{2}x_{A0})}$

$= \frac{2cD_{AB}}{\delta}\ln\left(\frac{1 - \frac{1}{2}(N_{Az}/k_1''c)}{1 - \frac{1}{2}x_{A0}}\right)$

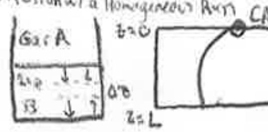
mb on spherical shell: $d/dr(r^2 N_{Ar}) = 0$

$N_{Ar} = x_A(N_{Ar} + N_{Br}) - cD_{AB}\frac{dx_A}{dr}$

$N_{Ar} = r^2\frac{-cD_{AB}}{1-x_A}\frac{dx_A}{dr}$

$W_A = 4\pi r^2 \cdot N_{Ar}|_{r_1}$

Diffusion w/ a Homogeneous Rxn



Given: A diffusion in B; Diffusion is isoT, undergoes irrev. 1st order rxn $A + B \to AB$

Pseudobinary assumption: ignore small amount of AB produced

$\Rightarrow x_A(N_{Az} + N_{Bz}) = 0$

Flux relationship: $N_{Az} = -cD_{AB}\frac{dx_A}{dz}$

Shell Balance: $\frac{dN_{Az}}{dz} + k_1''c_A = 0$

$\frac{d^2c_A}{dz^2} - \frac{k_1}{D_{AB}}c_A = 0$

B.C #1: $z = 0$, $c_A = c_{A0}$

B.C #2: $z = L$, $\frac{dc_A}{dz} = 0$

$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$

Dimensionalize: $\Gamma = c_A/c_{A0}$, $\zeta = z/L$, $\phi = \sqrt{\frac{k_1''L^2}{D_{AB}}}$

$\Rightarrow \frac{d^2\Gamma}{d\zeta^2} - \phi^2\Gamma = 0$ B.C: $\zeta = 0$, $\Gamma = 1$; $\zeta = 1$, $\frac{d\Gamma}{d\zeta} = 0$

Conc profile: $\Gamma = \frac{\cosh(\phi(1-\zeta))}{\cosh\phi}$

$\Rightarrow \frac{c_A}{c_{A0}} = \frac{\cosh\left[\sqrt{\frac{k_1''L^2}{D_{AB}}}(1 - z/L)\right]}{\cosh\left(\sqrt{\frac{k_1''L^2}{D_{AB}}}\right)}$

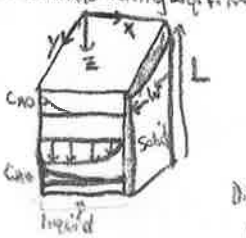
avg. conc: $\frac{c_{A,avg}}{c_{A0}} = \frac{\int_0^L (c_A/c_{A0})dz}{\int_0^L dz} = \frac{L\int_0^1 \Gamma d\zeta}{L\int_0^1 d\zeta} = \frac{\tanh\phi}{\phi}$

Flux: $N_{Az}|_{z=0} = -D_{AB}\frac{dc_A}{dz}\Big|_{z=0} = \left(\frac{c_{A0}D_{AB}}{L}\right)\phi\tanh\phi$

Fick's Law 1st: $J_A^* = -D_{AB}\nabla c_A$

2nd: $\frac{dc_A}{dt} = D_{AB}\nabla^2 c_A$

Diffusion into Falling Liq. Film



Assumptions: 'A' only slightly soluble in B; Laminar falling film; velocity field unaffected by diffusion

Diffusion takes place slowly: Penetration distance $\ll$ film thickness

Limiting solution $\to$ for short contact times

$x \ll \delta$, or $x/\delta \to 0$

Conc profile: $\frac{c_A}{c_{A0}} = \mathrm{erfc}\frac{x}{\sqrt{4D_{AB}z/V_{max}}}$

$V_z(x) = V_{max}\left[1 - (x/\delta)^2\right]$

Shell Balance: $\frac{dN_{Az}}{dz} + \frac{dN_{Ax}}{dx} = 0$

Flux relations: $N_{Az} = -c\cancel{D_{AB}\frac{dx_A}{dz}} + x_A(N_{Az} + N_{Bz})$

convection is driving force in z-dir

$N_{Ax} = -cD_{AB}\frac{dx_A}{dx} + \cancel{x_A(N_{Ax} + N_{Bx})}$

diffusion is driving force in x-dir

$V_{max}\left[1 - (x/\delta)^2\right]\frac{dc_A}{dz} = D_{AB}\frac{d^2c_A}{dx^2}$

B.C: $z = 0$, $c_A = 0$; $x = 0$, $c_A = c_{A0}$; $x = \delta$, $\frac{dc_A}{dx} = 0$

For short contact times

$V_{max}\frac{dc_A}{dz} = D_{AB}\frac{d^2c_A}{dx^2}$

B.C: $z = 0$, $c_A = 0$; $x = 0$, $c_A = c_{A0}$; $x \to \infty$, $c_A = 0$

$\Rightarrow \frac{d^2\phi}{d\eta^2} + 2\eta\frac{d\phi}{d\eta} = 0$

B.C: $\eta = 0$, $\phi = 1$; $\eta \to \infty$, $\phi = 0$

Combination of variables: $\phi = c_A/c_{A0}$, $\eta = \frac{x}{\sqrt{4D_{AB}z/V_{max}}}$

local flux: $N_{Ax}|_{x=0} = -D_{AB}\frac{dc_A}{dx}\Big|_{x=0} = c_{A0}\sqrt{\frac{D_{AB}V_{max}}{\pi z}}$

Total Molar Flow: $W_A = \int_0^W\int_0^L N_{Ax}|_{x=0}\,dx\,dy = WLc_{A0}\sqrt{\frac{4D_{AB}V_{max}}{\pi L}}$

For short contact time $(r/R)^2 \ll r/R$

Ch 19 $\phi = c_A$ Tubular Reactor

$V_z(r) = V_{max}(1 - (r/R)^2)$

$V_{zmax}(1 - (r/R)^2)\frac{dc_A}{dz} = D_{AB}\frac{1}{r}\frac{d}{dr}\left(r\frac{dc_A}{dr}\right)$

Let $y = R - r$

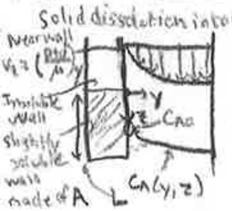
$2V_{zmax}\frac{y}{R}\frac{dc_A}{dz} = D_{AB}d^2c_A/dy^2$

$\eta = \frac{y}{R}\left(\frac{2V_{max}R^2}{9D_{AB}z}\right)^{1/3}$

BC: $z = 0$, $c_A = c_{A0}$; $y = 0$, $c_A = 0$; $y \to \infty$, $c_A = c_{A0}$

$\frac{c_A}{c_{A0}} = \frac{\int_0^\eta \exp(-\eta^3)d\eta}{\Gamma(4/3)}$

Solid dissolution into a falling liq film



Given: c_{A0} → Saturation conc. or solubility; A falling film fully developed flow; For $0 \le z \le L$, wall is made up of species A that is slightly soluble in B

$V_z = \frac{\rho g\delta^2}{2\mu}\left[1 - \left(1 - y/\delta\right)^2\right] = \frac{\rho g\delta^2}{2\mu}\left[2(y/\delta) - (y/\delta)^2\right]$

Shell Balance: $\frac{dN_{Ay}}{dy} + \frac{dN_{Az}}{dz} = 0$ neglect diffusion in z-dir; neglect convection in y-dir

Conc profile: $\frac{c_A}{c_{A0}} = \frac{\int_\eta^\infty \exp(-\eta^3)d\eta}{\int_0^\infty \exp(-\eta^3)d\eta}$ ← $\Gamma(4/3)$

$V_z = \left(\frac{\rho g\delta}{\mu}\right)y = ay$

$\frac{d^2\phi}{d\eta^2} + 3\eta^2\frac{d\phi}{d\eta} = 0$

$ay\frac{dc_A}{dz} = D_{AB}\frac{d^2c_A}{dy^2}$

$4/3\,\Gamma(4/3) = \Gamma(7/3)$

B.C: $z = 0$, $c_A = 0$; $y = 0$, $c_A = c_{A0}$; $y \to \infty$, $c_A = 0$

comb. of variables: $\phi = c_A/c_{A0}$, $\eta = y\left(a/9D_{AB}z\right)^{1/3}$

local flux: $N_{Ay} = -D_{AB}\frac{dc_A}{dy}\Big|_{y=0} = \frac{D_{AB}}{\Gamma(4/3)}\left(\frac{a}{9D_{AB}z}\right)^{1/3}$

Total molar flow: $W_A = \int_0^W\int_0^L N_{Ay}|_{y=0}\,dz\,dx = \frac{2D_{AB}c_{A0}WL}{\Gamma(7/3)}\left(\frac{a}{9D_{AB}L}\right)^{1/3}$

Thermal Penetration: $4\sqrt{\alpha t}$

Penetration depth: $4\sqrt{D_{AB} t}$

Damköhler # II: $Da_{II} = \frac{k_1 \delta^2}{D_{AB}}$

$\frac{d}{dx} \operatorname{erf} u = \frac{2}{\sqrt{\pi}} \exp(-u^2) \frac{du}{dx}$

$\frac{d}{dx}\left(\int_\eta^\infty \exp(-\bar{u}^3)\, d\bar{x}\right) = \exp(-u^3)\frac{du}{dx}$

Dimensionless Groups

$Re = [l v_0 \rho/\mu]$

$Pr = [\hat{C}_p \mu / k]$

$Gr = [g\beta(T_1 - T_0) l_0^3 / \nu^2]$

$Br = [\mu v_0^2 / k(T_1 - T_0)]$

$\hat{P}e = Re\, Pr$

$Ra = Gr\, Pr$

$Ec = Br/Pr$

$Nu = \frac{hl}{k}$

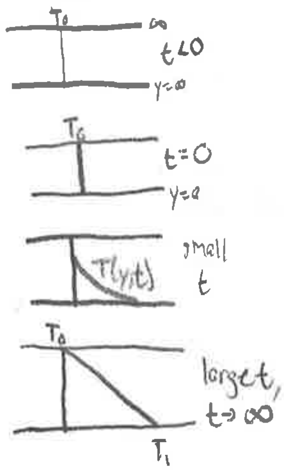
$Sh = \frac{k_x l_0}{c D_{AB}}$

$Sc = \mu/\rho D_{AB}$

$\Theta = \frac{T - T_0}{T_1 - T_0}$

Wall heat flux: $q|_{y=0} = -k \frac{dT}{dy}\Big|_{y=0} = \frac{k}{\sqrt{\pi \alpha t}}(T_1 - T_0)$

Heating of a Semi-Infinite Slab



$\frac{\partial T}{\partial t} = \alpha \nabla^2 T$

$\Rightarrow \frac{\partial \Theta}{\partial t} = \alpha \frac{\partial^2 \Theta}{\partial y^2}$

B.C.: $y=0$, $\Theta=1$, for all $t>0$

$y\to\infty$, $\Theta=0$ for all $t>0$

comb. of variables: $\eta = y/\sqrt{4\alpha t}$

$\Downarrow$

$\frac{d^2\Theta}{d\eta^2} + 2\eta \frac{d\Theta}{d\eta} = 0$

B.C.: $\eta=0$, $\Theta=1$

$\eta\to\infty$, $\Theta=0$

temp profile: $\frac{T - T_0}{T_1 - T_0} = \operatorname{erfc} \frac{y}{\sqrt{4\alpha t}}$

$= 1 - \operatorname{erf} \frac{y}{\sqrt{4\alpha t}}$

Heating of a Finite Slab

Define: $\Theta = \frac{T_1 - T}{T_1 - T_0}$

$\eta = y/b$, $\tau = \frac{\alpha t}{b^2}$

I.C: $t=0$, $T=T_0$

B.C: $y=\pm b$, $T=T_1$

$\Downarrow$

$\frac{\partial \Theta}{\partial \tau} = \frac{\partial^2 \Theta}{\partial \eta^2}$

I.C: $\tau=0$, $\Theta=1$ for $-1<\eta<1$

B.C: $\eta=\pm 1$, $\tau>0$

Sep. of variables

$\Theta(\eta,\tau) = f(\eta) g(\tau)$

plug back into ODE, divide by fg

$\frac{1}{g}\frac{dg}{d\tau} = \frac{1}{f}\frac{d^2 f}{d\eta^2} = -C^2$

Recall $\int \cos^2 u\, du = \frac{1}{2}u + \frac{1}{4}\sin 2u + C$

$\Rightarrow \frac{dg}{d\tau} = -C^2 g$, $\frac{d^2 f}{d\eta^2} = -C^2 f$

$\Rightarrow g = A\exp(-C^2\tau)$, $f = B\sin C\eta + C\cos C\eta$

b/c of symmetry about x-z plane we must have $f(\eta) = f(-\eta)$

since $\sin\Theta \neq \sin(-\Theta)$

but $\cos\Theta = \cos(-\Theta)$

use B.C at $\eta=\pm 1$, $\Theta=0$

$C\cos C = 0$, clearly $C \neq 0$

$\cos C = 0$ satisfied for $C_n = (n+\frac{1}{2})\pi$, $n=0,\pm1,\pm2,\ldots$ $B=0$

$\Rightarrow \Theta_n = A_n C_n \exp(-(n+\frac{1}{2})^2\pi^2\tau)\cos\left(\left(\frac{n+1}{2}\right)\pi\eta\right)$

b/c of linearity we can superpose all sol.

Also exp's cos form $\Rightarrow$ have same values as terms w/ $-(n+1)$

$D_n = A_n C_n + A_{-(n+1)} C_{-(n+1)}$

$\Theta = \sum_n \Theta_n = \sum_{n=0}^{\infty} D_n \exp[-(n+\frac{1}{2})^2\pi^2\tau]\cos[(n+\frac{1}{2})\pi\eta]$

$\Rightarrow$ mult. w/ $\cos(m+\frac{1}{2})\pi\eta$, int. from $\eta=-1$ to 1

using I.C.,

$1 = \sum_{n=0}^{\infty} D_n \cos(n+\frac{1}{2})\pi\eta \Rightarrow \int_{-1}^{1}\cos(m+\frac{1}{2})\pi\eta = \sum_{n=0}^{\infty} D_n \int_{-1}^{1}\cos(m+\frac{1}{2})\pi\eta\,\cos(n+\frac{1}{2})\pi\eta\, d\eta$

only terms w/ $m=n$ remain, $m\neq n \to 0$

$\Rightarrow \frac{\sin(m+\frac{1}{2})\pi\eta}{(m+\frac{1}{2})\pi}\Big|_{-1}^{1} = D_m \frac{\frac{1}{2}(m+\frac{1}{2})\pi\eta + \frac{1}{4}\sin 2(m+\frac{1}{2})\pi\eta}{(m+\frac{1}{2})\pi}\Big|_{-1}^{1} \Rightarrow D_m = \frac{2(-1)^m}{(m+\frac{1}{2})\pi}$

$$\Theta = \frac{T_1 - T}{T_1 - T_0} = 2\sum_{n=0}^{\infty} \frac{(-1)^n}{(n+\frac{1}{2})\pi}\exp\left(-(n+\frac{1}{2})^2\pi^2\alpha t/b^2\right)\cos(n+\frac{1}{2})\frac{\pi y}{b}$$

combined = molecular + convective

$\underline{n}_\alpha = \underline{j}_\alpha + \rho \underline{v} w_\alpha$

$\underline{\phi} = \underline{\pi} + \rho \underline{v}\underline{v}$

$\underline{e} = q + [\underline{\pi}\cdot\underline{v}] + \rho\underline{v}(\hat{U} + \frac{1}{2}v^2)$

Heat Transfer Coefficients:

$j_D = Sh\, Re^{-1} Sc^{-1/3}$ (mass transfer)

$j_H = Nu\, Re^{-1} Pr^{-1/3}$ (heat transfer)

Channel flow: $\frac{Nu_{ln}}{Re Pr^{1/3}} = j_{H,ln} \approx f/2$

Forced convection of submerged object: $j_{H,loc} = \frac{1}{2}f = .332\, Re^{-1/2}$ (Colburn analogy)

Flow around a sphere: $Nu_m = 2 + .6 Re^{1/2} Pr^{1/3}$

$Nu_m = 2 + (.4 Re^{1/2} + .06 Re^{2/3}) Pr^{.4}\left(\frac{\mu_\infty}{\mu_0}\right)^{1/4}$ (not valid for $Pr \to \infty$)

[Recommended $3.5 < Re < 7.6\times10^4$, $.71 < Pr < 380$, $1 < \mu_\infty/\mu_0 < 3.2$]

Flow around a cylinder: Whitakers

$Nu_m = (.4 Re^{1/2} + .06 Re^{2/3}) Pr^{.4}\left(\frac{\mu_\infty}{\mu_0}\right)^{1/4}$ [$1 < Re < 10^5$, $.67 < Pr < 300$, $.25 < \mu_\infty/\mu_0 < 5.2$]

McAdam

$Nu_m = (.376 Re^{1/2} + .057 Re^{2/3}) Pr^{1/3}$

Flow around other objects: $Nu_m - Nu_{m,0} = .6 Re^{1/2} Pr^{1/3}$

Forced convection thru packed beds: $j_H = 2.19 Re^{-2/3} + .78 Re^{-.38}$

for small Re, $j_H = 2.19 Re^{-2/3}$

Chilton Colburn Factor: $j_H = \frac{h_{loc}}{\hat{C}_p G_0}\left(\frac{\hat{C}_p\mu}{k}\right)^{2/3}$ w/ $Re = \frac{D_p G_0}{(1-\varepsilon)\mu\psi} = \frac{6 G_0}{a\mu\psi}$

effective particle size: $D_p = 6/a_v = \frac{6(1-\varepsilon)}{a}$

condensation: $h_m = .954\left(\frac{k^3 L\rho^2}{\mu w}\right)^{1/3}$

For mod. temp. diff.: $h_m = .725\left(\frac{k^3\rho^2 g\Delta\hat{H}_{vap}}{\mu D(T_d - T_0)}\right)^{1/4}$

Mass Transfer Coefficients

Falling film on plane surfaces: $Sh_m = \frac{k_{c,m} L}{D_{AB}} = 1.128(Re\, Sc)^{1/2}$

Solid dissolution: $Sh_m = \frac{k_{c,m} L}{D_{AB}} = 1.017\sqrt[3]{\frac{L}{\delta}}(Re\, Sc)^{1/3}$

Flow around spheres

$Sh_m = \frac{k^0_{c,m} D}{D_{AB}} = .6535(Re\, Sc)^{1/2}$ (gas absorption)

$Sh_m = k^0_{c,m} D/D_{AB} = .991(Re\, Sc)^{1/3}$ (solid sphere w/ slightly soluble coating)

Forced convection around spheres: $Sh_m = 2 + .6 Re^{1/2} Sc^{1/3}$

Convection along a flat plate: $j_{H,loc} = j_{D,loc} = \frac{1}{2}f_{loc} = .332 Re_x^{-1/2}$

$j_{H,loc} = \frac{Nu_{loc}}{Re Pr^{1/3}} = \frac{h_{loc}}{\rho\hat{C}_p v_\infty}\left(\frac{\hat{C}_p\mu}{k}\right)^{2/3}$

$j_{D,loc} = \frac{Sh_{loc}}{Re Sc^{1/3}} = \frac{k_{x,loc}}{c v_\infty}\left(\frac{\mu}{\rho D_{AB}}\right)^{2/3}$

Chilton-Colburn factors

Free convection

$\rho(T) = \bar{\rho} - \bar{\rho}\bar{\beta}(T - \bar{T})$

Stokes-Einstein

$D_{AB} = \frac{k_B T}{6\pi\mu_B r_A}$ size of solute